Transport Closure in Cosmological Inference and the Hubble Tension

A Certified Redshift-Active Analysis of Planck, DESI, Pantheon+, and SH0ES

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Abstract

Taking the publicly archived AASC dependency stack as established, we test whether the early-universe anchor and the SH0ES-scale late determination can be represented as one and the same cosmological parameter under a seven-coordinate transport-closure criterion. The primary theorem object promotes the correction to a three-coefficient redshift-active map that enters the retained Type Ia supernova and DESI BAO predictions while preserving the native Planck/CAMB early anchor.

The completed five-chain posterior uses the native Planck 2018 validation configuration, DESI DR1 BAO, and the calibrated Pantheon+SH0ES covariance, with the separate explicit SH0ES Gaussian summary factor removed. It gives

$$H_{0,\mathrm{RA}}^{\mathrm{AASC}} = 68.7319 \pm 0.3675 \ \mathrm{km\,s^{-1}\,Mpc^{-1}}$$

with

$$dT_{a0} = 0.0144 \pm 0.0590, \quad dT_{a1} = -0.0181 \pm 0.1063, \quad dT_{a2} = 0.0051 \pm 0.0669.$$

All three coefficients are measured in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, are statistically consistent with zero, and produce a posterior $\Delta H(z)$ band centered near zero across the retained late-sector redshift range. The maximum selected split-Rhat minus one is 0.000984. A matched unrestricted de-overlapped ΛCDM posterior gives $H_0 = 68.7185 \pm 0.3680 \ \mathrm{km\,s^{-1}\,Mpc^{-1}}$.

The redshift-active theorem object clears the restricted seven-coordinate AASC certificate. All three transport coefficients are prospectively typed as standing-bearing tensor witnesses, and the independent Jacobian/SVD audit gives

$$\operatorname{rank} J_\Pi = 6, \qquad \text{nullity } J_\Pi = 0, \qquad \operatorname{rank} J_{\Pi,\mathrm{transport}} = 3, \qquad C_M = 0.$$

Thus the admissible transport channel is explicitly constructed, observable-facing, and certificate-cleared, yet the posterior does not activate it at the SH0ES scale.

Conditional on the imported AASC results, the early anchor and the SH0ES-scale late determination therefore do not stand in the certified same-parameter transport relation inside the declared finite redshift-active object. This is not a rejection of SH0ES, not an evidence-based model-selection preference over ΛCDM, not a full Planck `clik/clipy` production result, and not a certificate over arbitrary transport functions or new-physics extensions outside the declared object.

Contents

1 Introduction

The Hubble tension—the discrepancy between early-universe determinations of the expansion rate and late-time distance-ladder measurements—remains one of the most persistent anomalies in contemporary cosmology. Planck analyses of the cosmic microwave background favor a present expansion rate near 67 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, while Cepheid-anchored Type Ia supernova calibrations favor a value near 73 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$.

Standard responses attribute the discrepancy either to unmodeled systematics or to additional cosmological dynamics. Both approaches presuppose that the compared early- and late-sector quantities are legitimate representations of one transported parameter. This manuscript tests that presupposition directly. The question is not merely whether a numerical fit can be improved, but whether a SH0ES-scale late value can be realized as a closed same-parameter transport of the early-universe anchor while the target, observable, calibration, continuation, estimator, and hidden-remainder ledgers remain fixed.

The formal analysis describes a seven-coordinate AASC residual object. The numerical analysis distinguishes that formal object from the executable penalty used by the retained chains. The primary numerical theorem object is redshift-active:

$$H_{\mathrm{AASC}}(z) = H_{\Lambda\mathrm{CDM}}(z) + \frac{dT_{a0} + dT_{a1} z/(1+z) + dT_{a2}[z/(1+z)]^2}{1 + (z/10)^4}.$$

All three coefficients enter the late-sector supernova and BAO predictions, while the early Planck/CAMB anchor remains fixed to the declared tolerance.

The primary posterior is a completed five-chain de-overlapped run using native Planck validation, DESI DR1, and calibrated Pantheon+SH0ES covariance, without the separate explicit SH0ES Gaussian summary factor. A matched unrestricted ΛCDM posterior is retained on the same observational ledger. The earlier certified present-epoch object is retained as a nested subcase, and the SH0ES-summary-inclusive AASC and ΛCDM runs are retained as archived-stack sensitivity comparisons.

The redshift-active posterior remains near $H_0 \simeq 68.73$, and all three transport coefficients remain consistent with zero. The result is not only multichain-supported. The finite theorem object also clears a restricted seven-coordinate certificate: the source, role, anchor, quotient, continuation, boundary, and estimator ledgers are fixed, and the declared covariance-whitened observable Jacobian is full rank with no hidden standing-bearing direction.

The resulting endpoint is precise. Within the certified redshift-active theorem object, the SH0ES-scale late value is not realized as the same admissibly transported expansion parameter from the early anchor. This does not imply that SH0ES is erroneous, does not establish an evidence preference over ΛCDM, and does not exclude a different theorem object with additional fields, transport functions, data ledgers, or dynamics.

The remainder of the paper fixes the AASC dependency structure, develops the formal residual map, distinguishes formal and executable objects, reports the redshift-active posterior and matched comparisons, presents the certificate, and states the exact claim boundary.

2 Dependency lock and division of labor

This manuscript is a downstream numerical specialization in the AASC series. It does not use the numerical posterior to establish the AASC kernel, the structure of the admissibility interface, or the cosmological divergence theorem. Those results are imported in a fixed dependency order, and the local burden of this paper begins only after that import. Table 1 records the public DOI for each dependency so that the upstream formal machinery is inspectable rather than private background.

Dependency source and public DOI	Imported result	Use in this manuscript
Non-Degenerate Construction and the Kernel of Admissibility [1] 10.5281/zenodo.19324199	Fixed-domain governance kernel $K = \{\mathrm{Adm}, \mathrm{St}, \mathrm{Ref}, \mathrm{Irr}\}$, governance equivalence, non-derivability, and no faithful same-domain counterexample	Fixes the target conditions under which same-domain construction, transport, and reportability are assessed.
The Structure of Admissibility [2] 10.5281/zenodo.19198249	AMetric proposal-side boundary, bivalent admissibility-relevant quotient, standing as reuse-stable admissibility, unique admissible interior, and conservation of standing	Excludes a second same-scope classifier, same-act repair, and undeclared boundary authority.
The Exhaustion Theorem [3] 10.5281/zenodo.18682423	Typed boundary discipline, fixed-scope exhaustion, and anchor/tensor/skin role decomposition	Supplies the finite-role and no-fourth-role discipline used to classify numerical changes.
Admissibility and Constraint-Induced Rigidity of Physical Quotient Structure [4] 10.5281/zenodo.18630901	AER observable-completeness and no-hidden-remainder specialization for physical quotient structure	Supplies the physical meaning of the hidden-remainder coordinate and the Jacobian null-space diagnostic.
On the Structural Origin of Cosmological Parameter Tensions and Transport-Induced Divergence [5] 10.5281/zenodo.19907827	External cosmological inference tuple, seven divergence loci, finite normal forms, and realization-certificate burden	Supplies the cosmology-specific transport theorem that is implemented and tested here.

Table 1: Formal dependency ledger with public DOI records. Imported results are not re-proved by the numerical posterior.

The logical order is therefore

$$\begin{aligned}\text{kernel} &\Longrightarrow \text{admissibility interface}\\ &\Longrightarrow \text{typed exhaustion and physical quotient discipline}\\ &\Longrightarrow \text{cosmological transport theorem}\\ &\Longrightarrow \text{retained executable inference.}\end{aligned}$$

No arrow is reversed. In particular, the empirical result does not serve as evidence for the primitive kernel theorem, and the fixed-domain closure claims are not inferred from the numerical location of the posterior.

The cosmology predecessor establishes the structural statement that a persistent early/late discrepancy under a same-target claim must have nonempty image in a finite family of anchor, tensor, quotient, continuation, boundary-record, estimator, and error-discipline loci. The present manuscript has the specific numerical role of specifying one executable transport representative, sampling its posterior, and auditing exactly which formal coordinates that representative does and does not activate.

Accordingly, two different standards of claim are maintained throughout. Statements about the zero set of the seven-coordinate residual map are formal statements about the imported and specialized theorem object. Statements about Planck, DESI, Pantheon+, SH0ES, the transport coefficients, or a penalty-weight scan are empirical statements about the retained executable representative. Section 10 makes that distinction explicit.

3 What follows when AASC is taken as established

Conditional on the imported kernel, admissibility-interface, exhaustion, AER-quotient, and cosmological transport-divergence results, the specialized residual map $\mathcal{F}^{\mathrm{form}}_{\mathrm{AASC}}$ and violation functional $\mathcal{V}^{\mathrm{form}}_{\mathrm{AASC}}$ do more than organize diagnostics. They define the fixed-domain criterion under which an early-to-late cosmological inference counts as transport of one and the same parameter-object.

Definition 3.1 (Certified same-parameter transport relation)**.** *Fix a finite cosmological theorem object $\mathcal{T}$. Write*

$$\theta_{\mathrm{E}} \sim^{\mathrm{AASC}}_{\mathcal{T}} \theta_{\mathrm{L}}$$

when there exists a realization $\mathcal{R} \in \mathcal{T}$ such that $\mathcal{F}^{\mathrm{form}}_{\mathrm{AASC}}(\mathcal{R}) = 0$, the declared transport map carries θ_{E} to θ_{L}, and the comparison is made in the fixed quotient-observable and error ledger of $\mathcal{T}$. This relation is scope-relative: changing the theorem object changes the relation being tested.

A completed same-domain transport therefore requires closure of all seven coordinates:

(1) anchor preservation;

(2) tensor-witness transport;

(3) quotient-observable closure;

(4) continuation consistency;

(5) boundary-record identity;

(6) estimator identity; and

(7) hidden-remainder closure, $C_M = 0$, so that no standing-bearing direction is invisible to the admitted observables.

The present paper constructs a finite theorem object with an explicit three-coefficient redshift-active transport map acting directly on the late-sector SN and BAO observables. It then runs the completed five-chain MCMC on the declared de-overlapped data ledger, clears the seven-coordinate certificate in the declared ledger/role/numerical sense, and performs an independent Jacobian/SVD audit. The full observable map is rank six with nullity zero, the transport subspace is rank three, and the hidden-remainder functional vanishes.

These facts establish that the channel itself is available: it is prospectively typed, observable-facing, and admissible within the declared object. The posterior nevertheless leaves all three transport coefficients statistically consistent with zero, keeps $H_{0,\mathrm{RA}}^{\mathrm{AASC}}$ near $68.73\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, and keeps $\Delta H(z)$ near zero across the retained SN/BAO range. The result is therefore stronger than a failure to locate an arbitrary numerical correction. Conditional on AASC, the early anchor and the SH0ES-scale late determination do not stand in the certified same-parameter transport relation defined above.

This conclusion is formal and scope-relative. It does not assert that early- and late-epoch expansion are unrestrictedly different physical phenomena; it states that the two determinations are not the same parameter under the admissible-transport criteria supplied by AASC for the certified redshift-active comparison object.

4 Operational dictionary for the numerical analysis

The terminology below distinguishes the formal seven-coordinate object, the executable penalty, and the primary redshift-active realization.

Anchor. The early-sector parameter block supplied by the native Planck/CAMB calculation. H_0, Ω_m, and r_{drag} enter the declared early-output ledger.

Redshift-active transport. The primary map $H_{\mathrm{AASC}}(z) = H_{\Lambda\mathrm{CDM}}(z) + \Delta H(z)$, with $\Delta H(z) = s(z)[dT_{a0} + dT_{a1}x + dT_{a2}x^2]$, $x = z/(1+z)$.

Transport witnesses. In the primary theorem object, $dT_{a0}, dT_{a1}, dT_{a2}$ are all data-active and standing-bearing tensor coordinates because they enter the SN and BAO predictions.

Present-epoch subcase. The nested earlier object in which only $H_0^{\mathrm{AASC}} = H_0 + dT_{a0}$ enters the late observables. It remains a certified consistency check, not the primary endpoint.

Formal violation functional. $\mathcal{V}^{\mathrm{form}}_{\mathrm{AASC}}$, assembled from anchor, tensor, quotient, continuation, boundary-record, estimator, and hidden-remainder residuals.

Executable penalty. $\mathcal{V}^{\mathrm{exec}}_{\mathrm{AASC}}$, the lower-dimensional five-term representative sampled with $\lambda = 10^6$. It is not pointwise identical to $\mathcal{V}^{\mathrm{form}}_{\mathrm{AASC}}$.

Hidden same-scope remainder. An observable-null direction that changes standing-bearing content. The certificate evaluates $C_M = \|J_S P_{\ker J_\Pi}\|^2_{\mathrm{HS}}$ independently of the fixed chain coordinate.

Primary de-overlapped stack. Native Planck validation, DESI DR1, and calibrated Pantheon+SH0ES covariance with the separate explicit SH0ES Gaussian summary factor removed.

Archived stack. The earlier likelihood ledger that additionally includes the explicit SH0ES Gaussian summary factor.

Matched baseline. An unrestricted fixed-transport ΛCDM run using the same de-overlapped observational ledger. It is a posterior-location comparator, not an evidence calculation.

Restricted certificate. The machine-readable (A, W, Q, C, B, E, M) certificate for the finite redshift-active theorem object. It is restricted to the declared map and ledger, not universal over arbitrary transport functions.

Constrained posterior. The posterior induced by the public likelihood stack plus $\lambda\mathcal{V}^{\mathrm{exec}}_{\mathrm{AASC}}$. The seven-coordinate certificate supplies the additional formal closure audit for the declared object.

5 Problem and contribution

Cosmological parameter tensions are usually approached numerically by enlarging a model class, fitting early and late data, and comparing likelihoods. In the AASC setting this is not the primitive numerical problem. A model can improve empirical fit while using different targets, different tensor witnesses, different calibration ledgers, different estimator rules, or hidden same-scope degrees of freedom. Such a fit is not a completed same-domain cosmological realization. It is an inference procedure with unclosed transport.

The problem addressed here is therefore:

> Which numerical cosmological realizations are admissible under the AASC constraints, and how is the admissible numerical solution space closed?

The answer is a constrained-inference theorem. Numerical analysis remains possible, but only after the solution space is restricted by AASC. Computation becomes search over admissible realizations rather than free fitting over arbitrary model extensions.

Main claim

Let $\mathcal{R}$ be a candidate numerical cosmological realization. Define the formal violation functional

$$\mathcal{V}^{\mathrm{form}}_{\mathrm{AASC}}(\mathcal{R}) = \alpha_A C_A + \alpha_T C_T + \alpha_Q C_Q + \alpha_C C_C + \alpha_B C_B + \alpha_E C_E + \alpha_M C_M,$$

with positive weights. The central result is:

> *The admissible numerical solution space is exactly the zero set of $\mathcal{V}^{\mathrm{form}}_{\mathrm{AASC}}$. Under regularity it is a constraint submanifold, and every non-skin numerical construction outside it belongs to a finite AASC normal-form failure space.*

This gives the numerical program:

$$\min_{\mathcal{R}} \ \chi^2_{\mathrm{E}}(\mathcal{R}) + \chi^2_{\mathrm{L}}(\mathcal{R}) + \chi^2_{\mathrm{aux}}(\mathcal{R}) \quad \text{subject to} \quad \mathcal{V}^{\mathrm{form}}_{\mathrm{AASC}}(\mathcal{R}) = 0.$$

Non-claims

The closure theorem itself does not introduce a new dark-energy dynamics and does not replace observational likelihood construction. It proves that same-domain numerical resolution of a parameter tension requires admissibility-preserving transport. The retained posterior is an empirical output of $\mathcal{V}^{\mathrm{exec}}_{\mathrm{AASC}}$, while the restricted certificate establishes all seven formal coordinates for the declared de-overlapped redshift-active theorem object. It does not certify arbitrary transport functions, parameterizations, or new-physics extensions outside the declared object. Independent early/late fitting, post-hoc calibration repair, hidden tensor variables, and estimator selection may remain useful diagnostics, but they are not completed same-domain resolutions unless their AASC residuals vanish.

6 AASC background used here

The paper imports the fixed-domain AASC kernel from [1], the AMetric/bivalent/standing-conservation interface from [2], typed exhaustion and anchor/tensor/skin discipline from [3], the AER observable-completeness specialization from [4], and the cosmological divergence ledger from

[5]. The dependency order is fixed in Section 2; none of these upstream results is inferred from the numerical posterior.

Only four consequences are used.

(A1) **No second same-scope classifier.** A same-domain construction cannot introduce an additional standing-bearing selector not fixed by the admissibility quotient.

(A2) **No same-act repair.** A failed or incomplete same-domain construction cannot be repaired post hoc as the same act.

(A3) **No hidden same-scope remainder.** Physically relevant structure invisible to all admitted observables is not standing-bearing content of the same fixed domain.

(A4) **Anchor/tensor/skin discipline.** Anchor data fix the domain; tensor data carry quotient-level standing content; skin changes no standing content.

These are invoked as prior theorems. The present paper specializes them to numerical cosmological inference.

7 External numerical cosmological inference tuple

The first task is to define the numerical object without AASC terminology. AASC classification is then applied to that external object.

Definition 7.1 (External numerical cosmological inference tuple)**.** *A numerical cosmological inference tuple is*

$$\mathcal{N} = (\mathcal{D}_{\mathrm{E}}, \mathcal{D}_{\mathrm{L}}, \mathcal{M}, \Theta, \Pi_{\mathrm{E}}, \Pi_{\mathrm{L}}, \widehat{\theta}_{\mathrm{E}}, \widehat{\theta}_{\mathrm{L}}, T, \Sigma, \mathcal{B}),$$

where:

(i) $\mathcal{D}_{\mathrm{E}}$ *is an early-epoch data model;*

(ii) $\mathcal{D}_{\mathrm{L}}$ *is a late-epoch data model;*

(iii) $\mathcal{M}$ *is a cosmological model class;*

(iv) Θ *is its parameter space;*

(v) $\Pi_{\mathrm{E}} : \Theta \to O_{\mathrm{E}}$ *and* $\Pi_{\mathrm{L}} : \Theta \to O_{\mathrm{L}}$ *are observable maps;*

(vi) $\widehat{\theta}_{\mathrm{E}} : \mathcal{D}_{\mathrm{E}} \to \Theta$ *and* $\widehat{\theta}_{\mathrm{L}} : \mathcal{D}_{\mathrm{L}} \to \Theta$ *are estimator or inference maps;*

(vii) $T = (T_A, T_W, T_Q, T_C, T_B, T_E)$ *is an early-to-late transport rule mapping early-inferred structure into late-inferred structure;*

(viii) Σ *is the error, covariance, tolerance, or likelihood discipline;*

(ix) $\mathcal{B} = (B_{\mathrm{E}}, B_{\mathrm{L}})$ *is the boundary-record and calibration ledger used by the data models.*

Definition 7.2 (Candidate numerical realization)**.** *A candidate numerical realization is a point*

$$\mathcal{R} = (\theta_G, \theta_T, \theta_B, T, \Pi_{\mathrm{E}}, \Pi_{\mathrm{L}}, \widehat{\theta}_{\mathrm{E}}, \widehat{\theta}_{\mathrm{L}})$$

inside an inference tuple $\mathcal{N}$*. Here* θ_G *denotes geometric or expansion variables,* θ_T *tensor-witness variables such as energy-content parameters or transport corrections, and* θ_B *boundary-record/calibration variables.*

Definition 7.3 (External numerical variation)**.** *An external numerical variation is a replacement of one or more components of $\mathcal{N}$ or $\mathcal{R}$ that changes the numerical inference procedure while preserving the claim that early and late outputs are estimates of one same-domain cosmological target. A variation is skin if it changes no data model, model class, parameter target, observable output, estimator rule, transport rule, error envelope, boundary record, or standing-bearing hidden direction.*

8 Same-domain type discipline

Definition 8.1 (Same-domain cosmological type)**.** *A candidate realization has same-domain type*

$$\tau_{\cos}(\mathcal{R}) = (A(\mathcal{R}), W(\mathcal{R}), Q(\mathcal{R}), C(\mathcal{R}), B(\mathcal{R}), E(\mathcal{R}), M(\mathcal{R})),$$

where:

- *A are anchor data: geometry, target expansion history, and model envelope;*
- *W are tensor witnesses: matter, radiation, dark energy, effective transport corrections, or other standing-bearing energy/field content;*
- *Q is the quotient-observable class: distance observables, spectra, expansion observables, and their equivalence relations;*
- *C is continuation structure: early-to-late evolution and transport consistency;*
- *B is the boundary-record ledger: calibration, records, and observational anchoring;*
- *E is the estimator/inference rule;*
- *M is the hidden-remainder coordinate: standing-bearing directions invisible to admitted observables.*

A transformation is same-domain type preserving iff it preserves all seven coordinates up to lawful equivalence.

Lemma 8.2 (Cosmological degrees-of-freedom lemma)**.** *Every admissibility-relevant numerical variation of an inference tuple $\mathcal{N}$ changes at least one of the seven coordinates A, W, Q, C, B, E, M. If it changes none, then it is skin relative to the cosmological inference type.*

Proof. The external tuple has exactly these possible numerical loci of standing-bearing change. A variation of the model envelope, geometry, or target expansion history changes A. A variation of matter/energy content, equation-of-state content, missing fields, effective corrections, or transport-correction coefficients changes W. A variation of observable maps or equivalence classes changes Q. A variation of early-to-late evolution, pipeline commutation, or transport changes C. A variation of calibration and record fixation changes B. A variation of estimator or likelihood projection changes E. A variation invisible to admitted observables but changing standing-bearing content changes M. If none of these changes, then all data models, targets, observables, estimators, records, continuations, error discipline, and standing-bearing degrees of freedom agree. By AASC role discipline the remaining difference is skin. $\square$

AASC role	Numerical cosmology coordinate	Typical observable meaning
Anchor	A	geometry, target expansion history, model envelope, fixed epoch target
Tensor witness	W	energy content, equation of state, transport correction, missing component
Quotient	Q	observable equivalence class: distances, spectra, expansion observables
Continuation	C	early-to-late transport, evolution rule, pipeline commutation
Boundary record	B	calibration, standards, record fixation, data anchoring
Estimator	E	inference map, likelihood projection, summary statistic
Hidden remainder	M	physical direction invisible to admitted observables

Table 2: AASC-to-numerical-cosmology mapping.

9 AASC residual map and admissible solution manifold

Definition 9.1 (AASC numerical residual map)**.** *For a candidate realization $\mathcal{R}$, define the residual map*

$$\mathcal{F}^{\mathrm{form}}_{\mathrm{AASC}}(\mathcal{R}) = (F_A, F_T, F_Q, F_C, F_B, F_E, F_M),$$

where

$$\begin{aligned}
F_A(\mathcal{R}) &:= A_{\mathrm{E}}(\mathcal{R}) - T_A A_{\mathrm{L}}(\mathcal{R}),\\
F_T(\mathcal{R}) &:= W_{\mathrm{E}}(\mathcal{R}) - T_W W_{\mathrm{L}}(\mathcal{R}),\\
F_Q(\mathcal{R}) &:= \Pi_{\mathrm{E}}(\mathcal{R}) - T_Q \Pi_{\mathrm{L}}(\mathcal{R}),\\
F_C(\mathcal{R}) &:= \mathcal{E}_{\mathrm{E}\to\mathrm{L}}(\theta_{\mathrm{E}}) - \theta_{\mathrm{L}},\\
F_B(\mathcal{R}) &:= B_{\mathrm{E}}(\mathcal{R}) - T_B B_{\mathrm{L}}(\mathcal{R}),\\
F_E(\mathcal{R}) &:= \widehat{\theta}_{\mathrm{E}}(\mathcal{D}_{\mathrm{E}}) - T_E \widehat{\theta}_{\mathrm{L}}(\mathcal{D}_{\mathrm{L}}),\\
F_M(\mathcal{R}) &:= J_S(\mathcal{R}) P_{\ker J_\Pi(\mathcal{R})}.
\end{aligned}$$

Here J_Π is the observable Jacobian and J_S the standing-content Jacobian defined in Section 13.

Definition 9.2 (AASC violation functional)**.** *The component residuals are*

$$\begin{aligned}
&C_A = \|F_A\|^2, && C_T = \|F_T\|^2, && C_Q = \|F_Q\|^2_{\Sigma^{-1}},\\
&C_C = \|F_C\|^2, && C_B = \|F_B\|^2, && C_E = \|F_E\|^2,\\
&C_M = \|F_M\|^2_{\mathrm{HS}}.
\end{aligned}$$

The total violation functional is

$$\mathcal{V}_{\mathrm{AASC}}^{\mathrm{form}}(\mathcal{R}) = \sum_{j \in \{A,T,Q,C,B,E,M\}} \alpha_j C_j(\mathcal{R}),$$

with $\alpha_j > 0$*.*

Definition 9.3 (Admissible numerical solution space)**.** *The admissible solution space is*

$$\mathcal{A}_{\mathrm{adm}} := \{\mathcal{R} : \mathcal{F}_{\mathrm{AASC}}^{\mathrm{form}}(\mathcal{R}) = 0\} = \{\mathcal{R} : \mathcal{V}_{\mathrm{AASC}}^{\mathrm{form}}(\mathcal{R}) = 0\}.$$

Theorem 9.4 (Violation-functional soundness and completeness)**.** *For a candidate realization* $\mathcal{R}$*,*

$$\mathcal{R} \in \mathcal{A}_{\mathrm{adm}} \iff \mathcal{R}$$

preserves anchors, tensor witnesses, quotient observables, continuation, boundary records, estimator identity, and observable completeness across the early and late inference regimes.

Proof. Because all weights are positive and every residual is nonnegative, $\mathcal{V}_{\mathrm{AASC}}^{\mathrm{form}}(\mathcal{R}) = 0$ iff every residual vanishes, which is equivalent to $\mathcal{F}_{\mathrm{AASC}}^{\mathrm{form}}(\mathcal{R}) = 0$. Vanishing of $F_A, F_T, F_Q, F_C, F_B, F_E, F_M$ is precisely preservation of the seven same-domain type coordinates. By the cosmological degrees-of-freedom lemma, any admissibility-relevant variation not preserving one of these coordinates is not same-domain type preserving. Conversely, preserving all seven coordinates leaves only skin-level differences. Hence the zero set of $\mathcal{V}_{\mathrm{AASC}}^{\mathrm{form}}$ is exactly the admissible solution space. □

Theorem 9.5 (Numerical closure theorem)**.** *Assume the realization space is a finite-dimensional smooth manifold* X*, the residual map* $\mathcal{F}_{\mathrm{AASC}}^{\mathrm{form}} : X \to Y$ *is continuous, and* $0 \in Y$ *is a regular value on the locus considered. Then:*

(i) $\mathcal{A}_{\mathrm{adm}} = (\mathcal{F}_{\mathrm{AASC}}^{\mathrm{form}})^{-1}(0)$ *is closed;*

(ii) on the regular locus $\mathcal{A}_{\mathrm{adm}}$ *is an embedded constraint submanifold of* X*;*

(iii) the tangent space is

$$T_{\mathcal{R}}\mathcal{A}_{\mathrm{adm}} = \ker d\mathcal{F}_{\mathrm{AASC},\mathcal{R}};$$

(iv) every non-skin numerical variation transverse to $\mathcal{A}_{\mathrm{adm}}$ *has a nonzero component in at least one of the seven residual coordinates and therefore realizes a finite AASC numerical normal form.*

Proof. Closedness follows from continuity of $\mathcal{F}_{\mathrm{AASC}}^{\mathrm{form}}$. The embedded-submanifold statement and tangent-space formula are the regular level-set theorem. If a variation is transverse to $\mathcal{A}_{\mathrm{adm}}$, then its differential under $\mathcal{F}_{\mathrm{AASC}}^{\mathrm{form}}$ is nonzero in at least one component. The seven components are exactly the same-domain type coordinates by construction and by the degrees-of-freedom lemma. Hence the variation realizes one of the finite normal forms described below. □

Corollary 9.6 (Nonemptiness as certificate condition)**.** *The admissible solution space is nonempty iff there exists at least one candidate realization carrying a complete AASC numerical certificate. In particular, nonemptiness is a certificate condition rather than a free-fitting assumption.*

Proof. A certificate with all residuals zero gives a point of $\mathcal{A}_{\mathrm{adm}}$. Conversely every point of $\mathcal{A}_{\mathrm{adm}}$ supplies the seven vanishing residuals, hence the certificate data recorded in Section 16. □

10 Formal residual object and executable representative

The formal closure theorem, the executable penalty, and the redshift-active certificate are related but non-identical objects.

Definition 10.1 (Formal transport-closure functional)**.** *The formal object is*

$$\mathcal{V}^{\mathrm{form}}_{\mathrm{AASC}}(\mathcal{R}) = \sum_{j\in\{A,T,Q,C,B,E,M\}} \alpha_j C_j(\mathcal{R}),$$

where the seven residual coordinates are those of Section 9. Its zero set is the formal admissible solution space $\mathcal{A}_{\mathrm{adm}}$*.*

Definition 10.2 (Retained executable representative)**.** *The v9 chains use*

$$\mathcal{V}^{\mathrm{exec}}_{\mathrm{AASC}}(\theta) = V^{\mathrm{exec}}_{\mathrm{anchor}} + V^{\mathrm{exec}}_{\mathrm{tensor}} + V^{\mathrm{exec}}_{\mathrm{quotient}} + V^{\mathrm{exec}}_{\mathrm{continuation}} + V^{\mathrm{exec}}_{\mathrm{AER\text{-}Max}},$$

implemented in `constraints/aasc_constraints.py`*. The sampled objective is*

$$-2\log\mathcal{L} = \chi^2_{\mathrm{public\ data}} + \lambda\mathcal{V}^{\mathrm{exec}}_{\mathrm{AASC}}, \qquad \lambda = 10^6.$$

The executable object is a declared finite representative. It is not pointwise identical to $\mathcal{V}^{\mathrm{form}}_{\mathrm{AASC}}$ *on the unrestricted formal realization space.*

Definition 10.3 (Redshift-active observable realization)**.** *In the primary theorem object, the executable transport coordinates also enter the late-sector observable map through*

$$H_{\mathrm{AASC}}(z) = H_{\Lambda\mathrm{CDM}}(z) + s(z)\big(dT_{a0} + dT_{a1}x + dT_{a2}x^2\big), \quad x = \frac{z}{1+z}.$$

Thus all three coefficients are observable-facing and standing-bearing in the redshift-active theorem object.

Proposition 10.4 (Claim-transfer boundary)**.** *A posterior statement obtained from* $\mathcal{V}^{\mathrm{exec}}_{\mathrm{AASC}}$ *licenses conclusions for the retained likelihood stack, transport map, priors, and executable terms. It does not by itself establish all seven formal residual coordinates. For the declared redshift-active object, that additional burden is met by the restricted certificate in Appendix D, including a full-rank covariance-whitened Jacobian/SVD audit.*

10.1 Implementation-faithfulness ledger

Formal coordinate	Executable representative	Chain/certificate status	Correct interpretation
F_A	Recombination-stop anchor term in the executable penalty	Executable term is basis-enforced; the actual redshift-active map is separately below the 10^{-6} recombination tolerance	Anchor preservation is certified by the declared source map and tolerance ledger, not by posterior variation alone.
F_T	Smoothness/roughness regularization over the coefficient basis	Nonzero support in the sampled penalty; all three coefficients also enter SN/BAO predictions	The penalty regularizes the transport sector, while tensor standing is established independently by prospective role typing and observable rank.
F_Q	Flat closure representative in $\mathcal{V}^{\mathrm{exec}}_{\mathrm{AASC}}$	Configuration-enforced in the penalty; physical quotient map is tested by covariance-whitened SN/BAO and anchor outputs	The certificate supplies the observable-map closure burden.
F_C	Broad coefficient-envelope threshold plus declared smooth $H_{\mathrm{AASC}}(z)$ map	Threshold not crossed; source validation checks zero limit, positivity, integration, and anchor suppression	Continuation is certified by the implemented redshift-active map and validation ledger.
F_B	No separate scalar penalty	Fixed by the de-overlapped Pantheon+SH0ES, DESI, and Planck ledger	Boundary identity is a declared ledger zero.
F_E	No separate scalar penalty	Fixed by Cobaya/CAMB, priors, likelihood classes, convergence rules, and post-processing	Estimator identity is a declared ledger zero.
F_M	Fixed hidden-remainder chain coordinate	Chain coordinate is not the test; the redshift-active Jacobian is rank six/nullity zero and gives $C_M = 0$	Hidden-remainder closure is supplied by the independent AER-Max audit.

Table 3: Formal-to-executable implementation-faithfulness ledger for the primary redshift-active theorem object.

The primary chain result and the certificate therefore perform different work. The chains determine the posterior location and transport amplitudes. The certificate closes the full declared (A, W, Q, C, B, E, M) object. Together they license the theorem-object result; neither licenses a claim about arbitrary transport functions or undeclared cosmological extensions.

11 Numerical objective and penalty convergence

Definition 11.1 (AASC-constrained cosmological objective)**.** *Let*

$$\chi^2_{\mathrm{data}}(\mathcal{R}) := \chi^2_{\mathrm{E}}(\mathcal{R}) + \chi^2_{\mathrm{L}}(\mathcal{R}) + \chi^2_{\mathrm{aux}}(\mathcal{R}).$$

The hard-constrained AASC problem is

$$\min_{\mathcal{R}\in\mathcal{A}_{\mathrm{adm}}} \chi^2_{\mathrm{data}}(\mathcal{R}).$$

The soft-penalty problem is

$$\min_{\mathcal{R}} \; J_\lambda(\mathcal{R}) := \chi^2_{\mathrm{data}}(\mathcal{R}) + \lambda\mathcal{V}^{\mathrm{form}}_{\mathrm{AASC}}(\mathcal{R}).$$

Theorem 11.2 (Numerical well-definedness and penalty convergence)**.** *Assume either compactness of the parameter domain or coercivity of J_λ, continuity of all data and residual terms, and nonemptiness of $\mathcal{A}_{\mathrm{adm}}$. Then the hard problem has a minimizer and each soft problem has a minimizer. If $\lambda_n \to \infty$ and $\mathcal{R}_n$ are minimizers of J_{λ_n} with a bounded subsequence, every accumulation point of that subsequence lies in $\mathcal{A}_{\mathrm{adm}}$ and minimizes the hard-constrained problem. If the hard minimizer is unique, the entire bounded minimizing subsequence converges to it.*

Proof. The zero set of the continuous violation functional is closed. Existence of hard and soft minimizers follows from Weierstrass under compactness, or from the direct method under coercivity and lower semicontinuity. For the penalty limit, fix any admissible competitor $\mathcal{R}_* \in \mathcal{A}_{\mathrm{adm}}$. Optimality gives

$$\chi^2_{\mathrm{data}}(\mathcal{R}_n) + \lambda_n \mathcal{V}^{\mathrm{form}}_{\mathrm{AASC}}(\mathcal{R}_n) \le \chi^2_{\mathrm{data}}(\mathcal{R}_*).$$

Thus $\lambda_n \mathcal{V}^{\mathrm{form}}_{\mathrm{AASC}}(\mathcal{R}_n)$ is bounded above and $\mathcal{V}^{\mathrm{form}}_{\mathrm{AASC}}(\mathcal{R}_n) \to 0$. Continuity gives admissibility of accumulation points. Comparing with any hard-admissible competitor and passing to the limit yields hard optimality. Uniqueness gives convergence of the bounded subsequence. □

Remark 11.3 (Tolerance envelopes)**.** *If a nonzero tolerance is used, the tolerance must be included in Σ and the boundary ledger B. The hard constraint is then zero residual relative to the declared envelope, not silent post-hoc relaxation.*

12 Residuals, normal forms, and numerical failure space

Definition 12.1 (Numerical divergence normal forms)**.** *A first non-skin numerical divergence realizes one of the following normal forms:*

N1 Anchor split. *The early and late targets or model envelopes differ.*

N2 Tensor omission. *A standing-bearing energy/content witness is omitted or inconsistently transported.*

N3 Quotient mismatch. *Observables are compared across inequivalent quotient classes.*

N4 Continuation patch. *Early-to-late continuation is repaired or patched post hoc.*

N5 Boundary repair. *Calibration or record data are adjusted as an external same-domain fix.*

N6 Estimator selector. *Distinct inference maps are selected without being part of the model type.*

N7 Hidden remainder. *A physical direction affects standing content while invisible to admitted observables.*

Theorem 12.2 (Residual-to-normal-form equivalence)**.** *At a candidate realization $\mathcal{R}$, the first nonzero residual coordinate determines the first numerical failure normal form according to Table 4. Conversely, each normal form produces a nonzero residual in the corresponding coordinate unless it has been explicitly promoted into the same-domain type ledger.*

Proof. Each residual is the norm of the difference between early and transported late values of one same-domain type coordinate. A first nonzero F_A means target or anchor drift, hence N1. A first nonzero F_T means tensor-witness mismatch, hence N2. A first nonzero F_Q means quotient-output noncommutation, hence N3. A first nonzero F_C means continuation failure, hence N4. A first nonzero F_B means boundary-record mismatch, hence N5. A first nonzero F_E means estimator nonidentity, hence N6. A first nonzero F_M means a standing-bearing direction invisible to observables, hence N7. Conversely, each named normal form changes the corresponding coordinate. If the change is incorporated into the type ledger and transported consistently, the residual vanishes; otherwise it is detected in that coordinate. ☐

Residual	Numerical manifestation	AASC normal form
F_A	early/late target, geometry, or model envelope differs	N1 anchor split
F_T	energy content or correction witness omitted/inconsistent	N2 tensor omission
F_Q	observables compared across inequivalent quotient classes	N3 quotient mismatch
F_C	early-to-late evolution or pipeline repaired after inference	N4 continuation patch
F_B	calibration or record ledger shifted after mismatch	N5 boundary repair
F_E	estimator/likelihood projection selected differently by epoch	N6 estimator selector
F_M	hidden standing direction invisible to admitted observables	N7 hidden remainder

Table 4: Residual-to-normal-form mapping for numerical cosmology.

Theorem 12.3 (Finite numerical normal-form theorem)**.** *Every numerical scheme that changes empirical fit while leaving the AASC zero set realizes at least one of N1–N7.*

Proof. By the degrees-of-freedom lemma, any non-skin change alters one of A, W, Q, C, B, E, M. By the residual-to-normal-form theorem, each coordinate change realizes the corresponding normal form. Therefore every non-skin improvement outside $\mathcal{A}_{\mathrm{adm}}$ realizes a normal form in the finite family. ☐

Theorem 12.4 (Closure of the admissible numerical solution manifold)**.** *No same-domain numerical inference scheme can eliminate a cosmological parameter tension by a non-skin change outside $\mathcal{A}_{\mathrm{adm}}$. Any such attempt either changes scope, imports auxiliary structure, performs post-hoc repair, introduces an estimator selector, or leaves a hidden same-scope remainder. If none of these occurs, the scheme is already an admissible constraint-native realization.*

Proof. Any non-skin attempt outside $\mathcal{A}_{\mathrm{adm}}$ has a normal-form image. N1 moves anchors or scope. N2 changes tensor content. N3 changes quotient identity. N4 is same-domain repair. N5 imports boundary authority. N6 adds a same-scope selector. N7 violates observable completeness. Each is ruled out as a same-domain admissible resolution by the imported AASC consequences summarized in Section 2. Conversely, if none occurs, all residuals vanish and the realization lies in $\mathcal{A}_{\mathrm{adm}}$. ☐

13 Hidden remainder and Fisher/Jacobian test

AER-Max in the numerical setting is tested by comparing observable sensitivity with standing-content sensitivity.

Definition 13.1 (Linear hidden remainder)**.** *Let $J_\Pi(\mathcal{R})$ be the Jacobian of admitted observables with respect to local coordinates on the realization space, and let $J_S(\mathcal{R})$ be the Jacobian of standing-bearing physical content. A linear hidden remainder exists at $\mathcal{R}$ iff there is a tangent vector v such that*

$$J_\Pi(\mathcal{R})v = 0, \qquad J_S(\mathcal{R})v \neq 0.$$

Define

$$C_M(\mathcal{R}) := \left\| J_S(\mathcal{R}) P_{\ker J_\Pi(\mathcal{R})} \right\|_{\mathrm{HS}}^2,$$

where $P_{\ker J_\Pi}$ is the orthogonal projection onto the observable null space and $\|\cdot\|_{\mathrm{HS}}$ is the Hilbert–Schmidt norm in a declared local chart.

Theorem 13.2 (Linear AER-Max test)**.** *At a regular numerical point $\mathcal{R}$, $C_M(\mathcal{R}) = 0$ iff every infinitesimal standing-bearing direction is detected by admitted observables. Thus $C_M = 0$ is the linearized observable-completeness condition for the declared maps and normalization.*

Proof. If $C_M = 0$, then $J_S v = 0$ for every $v \in \ker J_\Pi$, so no observable-invisible tangent direction changes standing content. Conversely, if no such hidden direction exists, then J_S vanishes on $\ker J_\Pi$, so the projected Hilbert–Schmidt norm is zero. □

Remark 13.3 (Redshift-active certificate result)**.** *The executable YAML still fixes `hidden_remainder_norm` to zero, so the stored chain coordinate is not the Jacobian test. The independent certificate uses the six-coordinate chart $(H_0, \Omega_m, r_{\mathrm{drag}}, dT_{a0}, dT_{a1}, dT_{a2})$ and a standing-content map in which all three transport coefficients are standing-bearing. The admitted-observable map combines the native Planck/CAMB anchor-output block with exact covariance-whitened DESI DR1 and calibrated Pantheon+SH0ES responses evaluated through the redshift-active source map. At the posterior mean, minimum-χ^2 point, and 24 retained posterior draws, the full declared Jacobian has rank six and nullity zero at both certificate thresholds. The transport subspace has rank three. Hence $P_{\ker J_\Pi} = 0$ and $C_M = 0$. Appendix D gives the complete audit.*

14 Tensor-correction ansatz and admissibility

At the formal level, a finite transport witness may be represented as

$$\Delta H(z) = \sum_{k=0}^{K} a_k \phi_k(z),$$

provided every standing-bearing coefficient is declared in the tensor ledger, preserves the early anchor, enters the admitted observable map, preserves continuation, and introduces no hidden remainder.

The primary realization uses

$$\Delta H(z) = \frac{dT_{a0} + dT_{a1}x + dT_{a2}x^2}{1 + (z/10)^4}, \qquad x = \frac{z}{1+z},$$

and applies it directly to the late-sector expansion and distance predictions. Consequently all three coefficients are observable-facing tensor witnesses. This differs from the nested present-epoch object, where only dT_{a0} entered the late observables. The difference is a declared theorem-object change, not a reclassification performed after inference.

Theorem 14.1 (Finite tensor-correction admissibility)**.** *A finite tensor-correction ansatz is an admissible same-domain numerical extension only if its standing-bearing coefficients enter the tensor ledger W, preserve anchor normalization, commute with quotient observables, preserve early-to-late continuation, and introduce no hidden remainder. If any condition fails, the correction realizes one of N1–N7 rather than an admissible solution.*

Proof. If a coefficient changes geometry or the target envelope it is N1; if it affects standing content while absent from the tensor ledger it is N2; if it changes the observable comparison class it is N3; if it patches continuation after inference it is N4; if it is absorbed into boundary records it is N5; if selected by an estimator rule it is N6; and if it carries standing content while invisible to admitted observables it is N7. Otherwise all residual coordinates vanish relative to the declared object. □

15 Two-epoch toy inference witness

The following toy model is not a cosmological dynamics. It is a witness that omission of a tensor coordinate produces parameter tension, and that completed transport removes it.

Definition 15.1 (Two-epoch toy inference model)**.** *Let the true standing content be $(h, u) \in \mathbb{R}^2$, where h is the expansion parameter and u is a tensor witness. Let early and late observed summaries be*

$$y_E = h + \beta u, \qquad y_L = h + \gamma u,$$

with $\beta \neq \gamma$. The incomplete model omits u and estimates $h_E := y_E$, $h_L := y_L$. The completed model includes u and imposes the single transport target (h, u).

Proposition 15.2 (Explicit mismatch under tensor omission)**.** *In the incomplete model,*

$$h_L - h_E = (\gamma - \beta)u.$$

Thus a nonzero omitted tensor witness produces an apparent expansion-parameter tension. In the completed model, both summaries are fitted by the same pair (h, u), and the residual continuation mismatch vanishes.

Proof. Substitution gives the displayed difference. When (h, u) is included, the equations $y_E = h+\beta u$ and $y_L = h + \gamma u$ are two constraints on the same transported standing content. The apparent separate estimates h_E, h_L are replaced by the single anchor h together with tensor witness u. Therefore the mismatch is not repaired after the fact; it is removed by completing the tensor ledger and preserving transport. □

16 Numerical realization certificate

Definition 16.1 (AASC numerical certificate)**.** *A numerical output is AASC-certified when it provides:*

(C1) parameter values and likelihood score;

(C2) anchor ledger and proof of anchor preservation;

(C3) tensor-witness ledger and transport map;

(C4) quotient-observable comparison class;

(C5) continuation/evolution map from early to late inference;

(C6) boundary-record/calibration ledger;

(C7) estimator/inference map declaration;

(C8) hidden-remainder/Jacobian test;

(C9) residual vector $(C_A, C_T, C_Q, C_C, C_B, C_E, C_M)$.

Theorem 16.2 (Realization-certificate theorem)**.** *A numerical cosmological result is an admissible same-domain resolution of a parameter tension iff it admits an AASC numerical certificate with all residuals zero, or within a declared nonzero tolerance envelope that is itself included as boundary data.*

Proof. If the certificate has zero residuals, the result lies in $\mathcal{A}_{\mathrm{adm}}$ by violation-functional soundness and completeness. If a tolerance envelope is declared, then the envelope is part of the boundary record and quotient comparison class; zero residuals are computed relative to that declared tolerance. Conversely, any admissible same-domain result must preserve all seven type coordinates and therefore can supply the listed certificate data. □

Remark 16.3 (Restricted certificate status)**.** *The publication archive supplies a complete seven-coordinate certificate for the declared de-overlapped redshift-active theorem object. The anchor, tensor, quotient, continuation, boundary, and estimator residuals are exact tolerance, role, map, or ledger zeros relative to that fixed object. The hidden-remainder residual is evaluated numerically and satisfies $C_M = 0$: the full declared Jacobian is rank six with nullity zero at the posterior mean, minimum-χ^2 point, and 24 retained posterior draws at both certificate thresholds. This certificate is finite and scope-relative; it does not extend automatically to arbitrary transport functions or enlarged cosmological state spaces.*

17 Relation to standard numerical practice

The framework is compatible with likelihood fitting, Bayesian inference, MCMC, nested sampling, Fisher analysis, and emulator-based cosmology. It changes the admissible model space, not the numerical tools. Standard tools remain valid after the admissible realization space has been specified.

- Separate early and late fits are diagnostic, not final same-domain resolutions, unless the transport residuals vanish.
- Calibration shifts are boundary-record moves and must be declared in B, not used as post-hoc repair.
- Extra fields or time-dependent dark-energy functions are tensor-witness additions and must be included in W, not hidden as fit parameters outside the ledger.
- Null directions of the observable Jacobian that change standing content violate AER-Max and are numerically testable; fixing a hidden-remainder coordinate to zero is not equivalent to performing that test.

18 Empirical redshift-active transport-closure inference

This section reports the primary redshift-active five-chain AASC posterior, its restricted seven-coordinate certificate, the matched unrestricted ΛCDM posterior on the same de-overlapped observational ledger, and the retained present-epoch and archived-stack comparisons. The primary redshift-active likelihood uses the native Planck 2018 validation configuration, DESI DR1 BAO, and the calibrated Pantheon+SH0ES covariance, with the separate explicit SH0ES Gaussian summary factor removed. The transport basis acts directly in the late-sector expansion and distance predictions across redshift.

18.1 Empirical claim-status ledger

Claim	Status	Scope
Formal zero set of $\mathcal{V}^{\text{form}}_{\text{AASC}}$ defines the admissible realization space	proved conditionally	Seven-coordinate same-domain residual object under the stated regularity assumptions.
Redshift-active AASC posterior is converged	supported	Five independent chains; 247,840 saved rows; 422,716 weighted samples; maximum selected split-$\widehat{R}-1 = 0.000984$.
All three transport coefficients are observable-facing tensor witnesses	supported	$dT_{a0}, dT_{a1}, dT_{a2}$ enter the retained redshift-active SN and DESI BAO predictions.
Restricted seven-coordinate certificate for the redshift-active object	supported	All seven coordinates pass; the declared Jacobian has rank six and nullity zero at every retained audit point and certificate threshold.
A SH0ES-scale redshift-active correction is activated	not supported	All three transport coefficients and the posterior $\Delta H(z)$ band remain statistically consistent with zero.
Matched de-overlapped unrestricted ΛCDM posterior is converged	supported	Five independent chains; retained as a posterior-location comparator, not an evidence calculation.
SH0ES is rejected or erroneous	outside claim	Quoted residuals are reference diagnostics, not independent rejection statistics.
All possible transport bases or new-physics extensions are excluded	outside claim	The certificate is restricted to the declared finite three-coefficient theorem object.

Table 5: Claim-status ledger for the primary redshift-active analysis.

18.2 Likelihood stacks and theorem-object hierarchy

The retained runs are organized as follows.

Primary redshift-active AASC.

Native Planck validation, DESI DR1, and calibrated Pantheon+SH0ES covariance, with `redshift_active_transport: true`, `include_shoes: false`, and $\lambda = 10^6$. This is the primary empirical and certified theorem object.

Matched de-overlapped ΛCDM.
The same observational ledger, with transport fixed to zero and $\lambda = 0$. This is the matched posterior-location comparator.

Certified present-epoch subcase.
The earlier de-overlapped five-chain AASC object in which only $H_0^{\mathrm{AASC}} = H_0 + dT_{a0}$ enters the late observables. It is retained as a nested consistency check.

Archived-stack comparisons.
AASC and matched unrestricted ΛCDM posteriors that also include the separate explicit SH0ES Gaussian summary factor. They are retained as sensitivity and provenance results.

The redshift-active theorem object is finite and explicit:

$$\mathcal{T}_{\mathrm{RA}}^{\mathrm{AASC}} = (\mathcal{D}_{\mathrm{Planck}}^{\mathrm{native}}, \mathcal{D}_{\mathrm{DESI}}, \mathcal{D}_{\mathrm{P+SH0ES}}^{\mathrm{cov}}, \Theta_{\mathrm{RA}}, T_{\mathrm{RA}}, \Pi_{\mathrm{RA}}, \Sigma_{\mathrm{deov}}, E_{\mathrm{Cobaya}}). \tag{1}$$

The Planck/CAMB sector supplies the early anchor; the redshift-active map modifies only the declared late-sector observable construction.

18.3 Redshift-active transport implementation

The primary transport map is

$$\begin{aligned} H_{\mathrm{AASC}}(z) &= H_{\Lambda\mathrm{CDM}}(z) + \Delta H(z), \\ \Delta H(z) &= s(z)\left[dT_{a0} + dT_{a1}x + dT_{a2}x^2\right], \qquad x = \frac{z}{1+z}, \\ s(z) &= \frac{1}{1+(z/z_{\mathrm{stop}})^p}, \qquad z_{\mathrm{stop}} = 10, \quad p = 4. \end{aligned} \tag{2}$$

At the present epoch,

$$H_{0,\mathrm{RA}}^{\mathrm{AASC}} = H_0 + dT_{a0}, \tag{3}$$

because $x(0) = 0$ and $s(0) = 1$. At nonzero redshift, all three coefficients enter the late-sector predictions for $H(z)$, $D_H = c/H$, D_M, D_L, the SN distance modulus, and the DESI observables D_M/r_d, D_H/r_d, and D_V/r_d where applicable. Consequently $dT_{a0}, dT_{a1}, dT_{a2}$ are prospectively typed as standing-bearing tensor coordinates in $\mathcal{T}_{\mathrm{RA}}^{\mathrm{AASC}}$.

The damping function keeps the correction in the late-sector transport role. Across the declared prior corners, the maximum correction at recombination is $3.89 \times 10^{-7}\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, below the certificate tolerance $10^{-6}\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$. The source validation also checks the zero-correction ΛCDM limit, independent shape effects from dT_{a1} and dT_{a2}, positivity of $H_{\mathrm{AASC}}(z)$, numerical stability of the distance integrals, and the de-overlapped configuration flags.

18.4 Primary redshift-active five-chain posterior

The primary ensemble contains five Cobaya chain files, 247,840 saved rows, and 422,716 weighted samples. The selected-parameter split diagnostic satisfies

$$\max(\widehat{R} - 1) = 0.000984, \tag{4}$$

with the maximum occurring for Ω_m. This is below the target threshold $\widehat{R} - 1 < 0.01$.

Parameter	Mean	Std. dev.	68% interval	95% interval
$H_{0,\mathrm{RA}}^{\mathrm{AASC}}$	68.7319	0.3675	[68.3682, 69.0973]	[68.0129, 69.4556]
dT_{a0}	0.0144	0.0590	[-0.0440, 0.0730]	[-0.1006, 0.1310]
dT_{a1}	-0.0181	0.1063	[-0.1248, 0.0871]	[-0.2265, 0.1908]
dT_{a2}	0.0051	0.0669	[-0.0614, 0.0716]	[-0.1254, 0.1354]
$V_{\mathrm{AASC}}^{\mathrm{exec}}$	3.064×10^{-6}	2.497×10^{-6}	[8.47×10^{-7}, 5.26×10^{-6}]	[2.23×10^{-7}, 9.55×10^{-6}]
Ω_m	0.297528	0.004648	[0.292915, 0.302126]	[0.288452, 0.306701]
r_{drag} [Mpc]	147.537	0.218	[147.322, 147.754]	[147.110, 147.965]

Table 6: Primary redshift-active five-chain AASC posterior. The ensemble contains 247,840 saved rows and 422,716 weighted samples.

The principal result is

$$H_{0,\mathrm{RA}}^{\mathrm{AASC}} = 68.7319 \pm 0.3675 \ \mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{5}$$

The redshift-active coefficients are

$$dT_{a0} = 0.0144 \pm 0.0590, \tag{6}$$
$$dT_{a1} = -0.0181 \pm 0.1063, \tag{7}$$
$$dT_{a2} = 0.0051 \pm 0.0669, \tag{8}$$

in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$. Each is statistically consistent with zero. The displacement from the posterior mean to the SH0ES central value is

$$73.04 - 68.7319 = 4.3081 \ \mathrm{km\,s^{-1}\,Mpc^{-1}},$$

far larger than the inferred transport amplitudes.

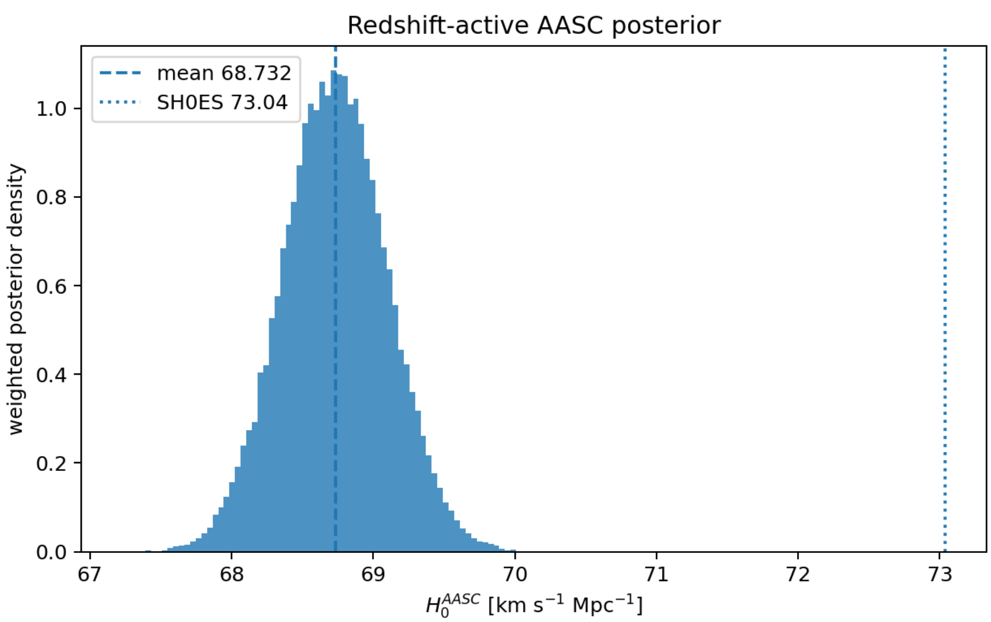
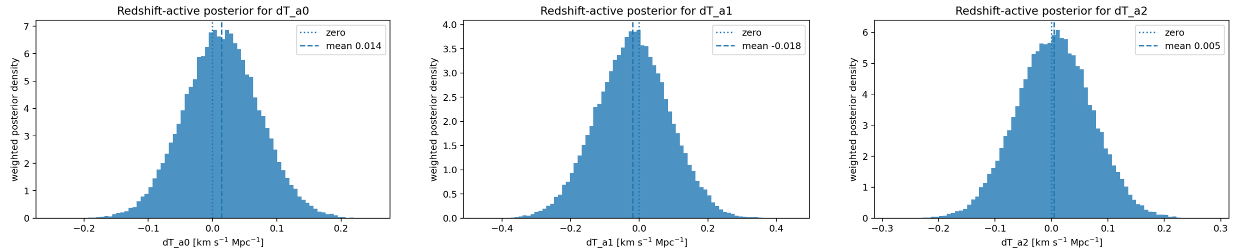


Figure 1: Primary redshift-active five-chain posterior for $H_{0,\mathrm{RA}}^{\mathrm{AASC}}$. The posterior remains in the Planck/BAO-supported region rather than migrating to the SH0ES central value.



(a) dT_{a0} (b) dT_{a1} (c) dT_{a2}

Figure 2: Marginal posteriors for the three standing-bearing redshift-active transport coefficients. All are statistically consistent with zero.

18.5 Transport evolution across redshift

The posterior transport correction remains centered near zero throughout the redshift range probed by the retained SN and BAO data.

z	mean $\Delta H(z)$	std. dev.	68% interval	95% interval
0.0	0.01437	0.05897	[-0.04399, 0.07297]	[-0.10061, 0.13099]
0.1	0.01276	0.05136	[-0.03810, 0.06369]	[-0.08777, 0.11436]
0.5	0.00888	0.03550	[-0.02612, 0.04428]	[-0.06119, 0.07832]
1.0	0.00656	0.02737	[-0.02041, 0.03404]	[-0.04731, 0.05960]
2.0	0.00451	0.02033	[-0.01554, 0.02488]	[-0.03581, 0.04400]
5.0	0.00259	0.01316	[-0.01048, 0.01565]	[-0.02341, 0.02821]

Table 7: Posterior summary of the redshift-active correction $\Delta H(z)$, in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$.

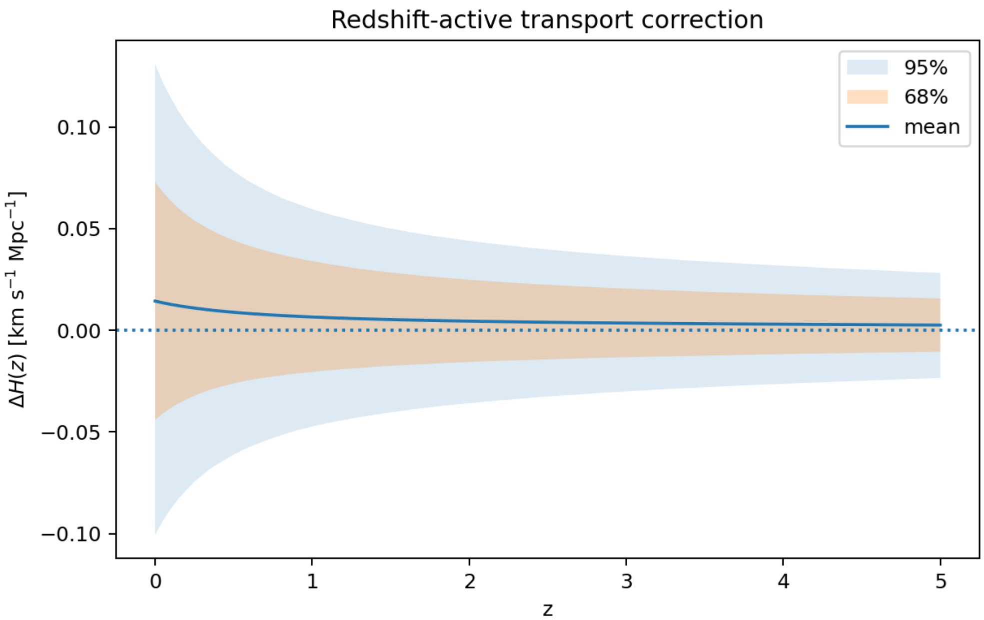


Figure 3: Posterior band for the redshift-active correction $\Delta H(z)$. The band remains concentrated near zero across the retained late-sector redshift range.

18.6 Restricted seven-coordinate certificate for the redshift-active object

The publication archive supplies the complete restricted certificate for the finite theorem object in Eq. (1). The certificate coordinate chart is

$$(H_0, \Omega_m, r_{\mathrm{drag}}, dT_{a0}, dT_{a1}, dT_{a2}),$$

and the standing-content chart is

$$(H_{0,\mathrm{RA}}^{\mathrm{AASC}}, dT_{a0}, dT_{a1}, dT_{a2}, \Omega_m, r_{\mathrm{drag}}).$$

All three transport coefficients are standing-bearing. The admitted observable map combines the retained Planck/CAMB anchor-output block, exact covariance-whitened DESI DR1 BAO residuals, and exact covariance-whitened calibrated Pantheon+SH0ES residuals with analytic absolute-magnitude marginalization.

At the posterior mean, the minimum-χ^2 sample, and 24 deterministic weighted posterior draws, the full declared Jacobian is rank six with nullity zero at both relative singular-value thresholds 10^{-6} and 10^{-5}. The three-dimensional transport subspace has rank three at every retained audit point. Therefore

$$P_{\ker J_\Pi} = 0, \qquad C_M = \|J_S P_{\ker J_\Pi}\|_{\mathrm{HS}}^2 = 0.$$

The late-sector-only map is also full rank at the declared certificate thresholds; its larger condition number and a deliberately stricter 10^{-4} stress threshold are retained as conditioning diagnostics rather than used to redefine the certificate. The resulting seven-coordinate residual vector is

$$(C_A, C_T, C_Q, C_C, C_B, C_E, C_M) = (0, 0, 0, 0, 0, 0, 0) \tag{9}$$

relative to the declared ledger modes and numerical tolerances. Appendix D gives the complete method and claim boundary.

18.7 Matched baseline and nested present-epoch comparison

The matched unrestricted de-overlapped ΛCDM posterior gives

$$H_0^{\Lambda\mathrm{CDM}} = 68.7185 \pm 0.3680\ \mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{10}$$

The redshift-active AASC and matched ΛCDM posterior means differ by only $0.0134\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, negligible compared with either uncertainty. The baseline is a posterior-location comparator; no evidence or model-selection preference is inferred.

The certified present-epoch subcase gives

$$H_{0,\mathrm{PE}}^{\mathrm{AASC}} = 68.7382 \pm 0.3618, \qquad \Delta_T(0) = 0.0241 \pm 0.0590.$$

Promotion of the transport basis into redshift-dependent SN and BAO observables therefore leaves the posterior location essentially unchanged while subjecting dT_{a1} and dT_{a2} to direct observable leverage and certificate testing.

For provenance, the archived-stack results are

$$H_{0,\mathrm{archived}}^{\mathrm{AASC}} = 69.2078 \pm 0.3519, \qquad H_{0,\mathrm{archived}}^{\Lambda\mathrm{CDM}} = 69.2156 \pm 0.3456.$$

The separate explicit SH0ES Gaussian factor shifts both archived posterior means upward by about $0.5\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ without activating the AASC transport sector.

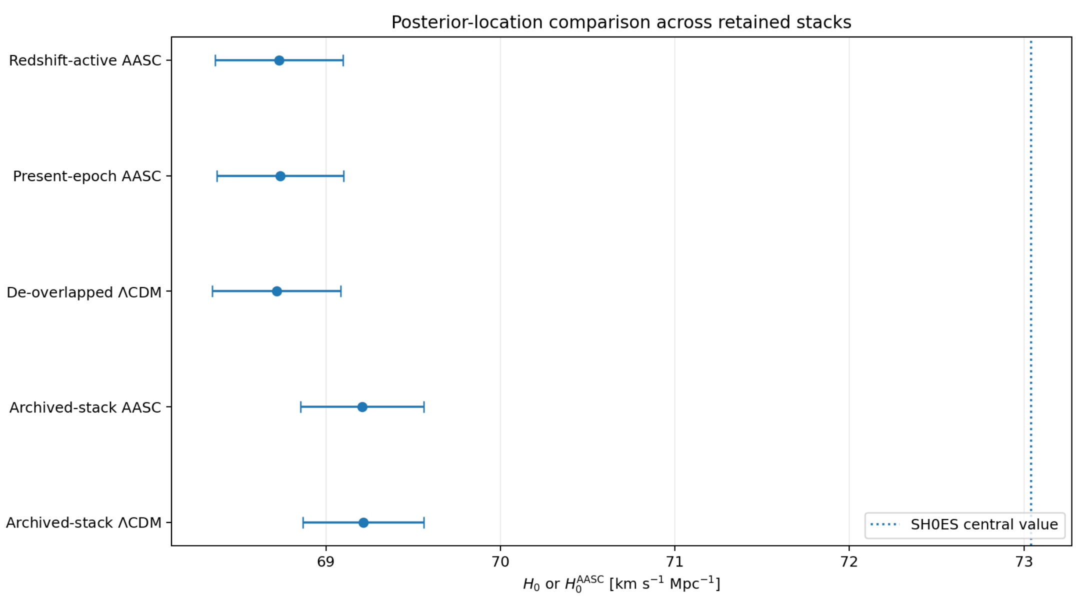


Figure 4: Posterior-location comparison across the primary redshift-active object, the present-epoch subcase, the matched de-overlapped ΛCDM baseline, and the archived-stack sensitivity runs. Error bars show one posterior standard deviation.

18.8 Likelihood and penalty budget

For the primary redshift-active posterior, the posterior-weighted mean contributions are

Component	Posterior-weighted mean χ^2
Native Planck validation block	1023.834
Pantheon+SH0ES calibrated covariance	1553.378
DESI DR1 BAO	13.482
Explicit SH0ES Gaussian summary	0.000
Executable AASC penalty $10^6\mathcal{V}^{\rm exec}_{\rm AASC}$	3.064
Total	2593.759

Table 8: Posterior-weighted likelihood and penalty budget for the primary redshift-active ensemble. The table is a diagnostic decomposition of the sampled objective, not an evidence calculation.

The executable penalty retains the same lower-dimensional constraint implementation as the present-epoch run. The formal certificate does not follow from the small scalar penalty alone; it follows from the separate role, ledger, observable-map, and Jacobian/SVD audits.

18.9 Comparison with the SH0ES reference

For $H_0^{\rm SH0ES} = 73.04 \pm 1.04\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, the primary reference distance is approximately 3.91σ. The corresponding calculation is

$$\frac{73.04 - 68.7319}{\sqrt{1.04^2 + 0.3675^2}} \simeq 3.91. \tag{11}$$

This is a reference-distance diagnostic, not an independent rejection statistic. The separate SH0ES Gaussian summary factor is excluded from the primary likelihood, while the calibrated Pantheon+SH0ES covariance remains part of the late-sector data ledger.

18.10 Penalty, replication, and claim boundary

The retained $\lambda = 10^5, 10^6, 10^7$ scan belongs to the archived present-epoch stack and remains a provenance sensitivity diagnostic. A full redshift-active multichain λ-profile is not claimed. Nor does this paper report Bayesian evidence, a full Planck `clik/clipy` production replication, or exhaustion of transport bases outside the declared three-coefficient map.

The exact supported statement is stronger than the earlier present-epoch endpoint but remains scope-relative: the source-supported, five-chain redshift-active theorem object clears the restricted seven-coordinate AASC certificate, and no SH0ES-scale same-parameter transport correction is activated. The result is not a systematic-error diagnosis, a rejection of SH0ES, a model-selection victory over ΛCDM, or an exclusion of richer dynamical objects.

18.11 Empirical conclusion

The completed redshift-active posterior gives

$$H_{0,\rm RA}^{\rm AASC} = 68.7319 \pm 0.3675\ \mathrm{km\,s^{-1}\,Mpc^{-1}},$$

with $dT_{a0}, dT_{a1}, dT_{a2}$ all consistent with zero and with $\Delta H(z)$ centered near zero across the retained SN/BAO redshift range. The full declared Jacobian is rank six with nullity zero, the transport subspace is rank three, and the seven-coordinate residual vector clears. The result is therefore not merely a failure to find an arbitrary numerical bridge: the declared channel is admissible, observable-facing, and certificate-cleared, yet the posterior does not activate it at the SH0ES scale.

Conditional on the imported AASC results, the early anchor and the SH0ES-scale late determination do not stand in the certified same-parameter transport relation within the declared finite theorem object.

19 Certified failure of SH0ES-scale same-parameter transport

The empirical and certificate layers establish that the declared transport channel is available in the precise AASC sense: it is explicitly parameterized, prospectively role-typed, coupled to the retained SN and BAO observables, and closed by the seven-coordinate certificate. The remaining question is therefore not whether an admissible channel was supplied, but whether the posterior occupies that channel at the amplitude required to identify the SH0ES-scale late determination with the early anchor.

Theorem 19.1 (Failure of SH0ES-scale same-parameter transport)**.** *Assume the imported AASC dependency stack and let $\mathcal{T}_{\mathrm{RA}}^{\mathrm{AASC}}$ denote the declared de-overlapped redshift-active theorem object of Eq.* (1)*. Suppose its restricted certificate clears,*

$$(C_A, C_T, C_Q, C_C, C_B, C_E, C_M) = (0, 0, 0, 0, 0, 0, 0),$$

relative to the declared ledger modes and numerical tolerances, and suppose the retained five-chain posterior is summarized by

$$H_{0,\mathrm{RA}}^{\mathrm{AASC}} = 68.7319 \pm 0.3675 \ \mathrm{km\,s^{-1}\,Mpc^{-1}}$$

and

$$dT_{a0} = 0.0144 \pm 0.0590, \quad dT_{a1} = -0.0181 \pm 0.1063, \quad dT_{a2} = 0.0051 \pm 0.0669.$$

Then the early anchor and the SH0ES-scale late determination do not stand in the certified AASC same-parameter transport relation within $\mathcal{T}_{\mathrm{RA}}^{\mathrm{AASC}}$.

Proof. Section 3 defines same-parameter transport within a fixed theorem object by the existence of a realization that both closes $\mathcal{F}_{\mathrm{AASC}}^{\mathrm{form}}$ and carries the early parameter-object to the late parameter-object through the declared transport map. The certificate verifies the first requirement for $\mathcal{T}_{\mathrm{RA}}^{\mathrm{AASC}}$: the anchor, tensor, quotient, continuation, boundary, and estimator coordinates close, while the observable Jacobian is rank six with nullity zero, the transport subspace is rank three, and $C_M = 0$. The channel is therefore admissible, observable-facing, and free of an unrepresented standing-bearing route.

The second requirement is not realized at the SH0ES scale. The posterior places all three transport coefficients near zero, and the induced $\Delta H(z)$ band remains centered near zero throughout the retained late-sector redshift range. At the present epoch, the required displacement is

$$73.04 - 68.7319 = 4.3081 \ \mathrm{km\,s^{-1}\,Mpc^{-1}},$$

whereas the reported 95% interval for $dT_{a0} = \Delta H(0)$ is $[-0.10061, 0.13099]$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$. The redshift-shape coefficients are likewise centered near zero and do not generate the required displacement at any retained SN/BAO redshift. Because the full transport chart is observable-facing and $C_M = 0$, no hidden same-scope standing-bearing route remains within the declared object. Hence the SH0ES-scale late determination is not the same transported parameter-object as the early anchor under the certified AASC relation for $\mathcal{T}_{\mathrm{RA}}^{\mathrm{AASC}}$. □

Corollary 19.2 (Conditions required for same-parameter identification)**.** *Within the AASC transport framework, identification of the early anchor with a SH0ES-scale late determination would require at least one of the following inside a certified theorem object:*

(i) a transport posterior that realizes the SH0ES-scale displacement while all seven certificate coordinates remain closed;

(ii) a richer transport basis, modified functional form, or additional tensor witness that clears the same certificate and realizes the required displacement; or

(iii) an explicitly revised data, observable, quotient, boundary-record, or estimator ledger under which the SH0ES-scale value becomes admissible in a newly declared comparison object.

None of these occurs in $\mathcal{T}_{\mathrm{RA}}^{\mathrm{AASC}}$. The certificate is clean, the declared transport subspace is fully observable, and the posterior transport remains negligible. The two determinations therefore remain formally distinct within this object.

Remark 19.3 (Interpretive boundary)**.** *The theorem is a conditional same-parameter result, not an unrestricted ontological claim that the universe contains two unrelated expansion phenomena. It establishes that the early anchor and the SH0ES-scale late determination fail the formal same-domain transport relation supplied by the imported AASC framework for the declared finite theorem object.*

20 Conclusion

Taking the public AASC dependency stack as established, the formal residual map $\mathcal{F}_{\mathrm{AASC}}^{\mathrm{form}}$ and violation functional $\mathcal{V}_{\mathrm{AASC}}^{\mathrm{form}}$ supply a precise criterion for when early- and late-sector inferences concern one and the same transported cosmological parameter. A completed same-domain transport must preserve the anchor, tensor-witness, quotient-observable, continuation, boundary-record, estimator, and hidden-remainder coordinates.

The paper constructs that channel explicitly. The primary theorem object lets three standing-bearing transport coefficients modify the late-sector expansion, Type Ia supernova distances, and DESI BAO observables while preserving the native Planck/CAMB early anchor. The completed five-chain posterior gives

$$H_{0,\mathrm{RA}}^{\mathrm{AASC}} = 68.7319 \pm 0.3675 \ \mathrm{km\,s^{-1}\,Mpc^{-1}},$$

with

$$dT_{a0} = 0.0144 \pm 0.0590, \quad dT_{a1} = -0.0181 \pm 0.1063, \quad dT_{a2} = 0.0051 \pm 0.0669.$$

All three coefficients are statistically consistent with zero, and the posterior $\Delta H(z)$ band remains centered near zero across the retained late-sector redshift range. The matched unrestricted de-overlapped ΛCDM posterior gives $H_0 = 68.7185 \pm 0.3680 \ \mathrm{km\,s^{-1}\,Mpc^{-1}}$.

The restricted certificate clears all seven coordinates for the declared finite object. The full covariance-whitened observable Jacobian is rank six with nullity zero at the posterior mean, minimum-χ^2 point, and 24 retained posterior draws at both certificate thresholds. The transport subspace is rank three, so $C_M = 0$. Together with the exact ledger and map conditions,

$$(C_A, C_T, C_Q, C_C, C_B, C_E, C_M) = (0,0,0,0,0,0,0).$$

The result therefore does not arise because the channel is ill-typed, observationally inert, or concealing a standing-bearing null direction. The admissible channel is present and audited; the posterior does not activate it at the SH0ES scale.

Theorem 19.1 gives the strengthened endpoint. Conditional on AASC, the early anchor and the SH0ES-scale late determination do not stand in the certified same-parameter transport relation within the declared redshift-active object. The late value remains formally distinct from the early anchor as a matter of certified same-domain structure, not merely as an unexplained numerical discrepancy. Corollary 19.2 states what would be required to change that status: a posterior-supported certified transport to the SH0ES scale, a richer certificate-clearing transport/tensor object that realizes the displacement, or an explicit revision of the comparison ledger.

The scope remains exact. The result does not declare SH0ES erroneous, does not establish an evidence preference against unrestricted ΛCDM, does not represent the native Planck validation route as a full `clik/clipy` production calculation, and does not exhaust arbitrary transport functions or richer new-physics theorem objects. Such alternatives require newly declared source maps, ledgers, posteriors, and certificates.

The publication archive contains the public formal dependency bundle, the redshift-active source implementation and validation, all retained posterior comparisons, exact run metadata and chain archives, the machine-readable seven-coordinate certificate, Jacobian/SVD outputs, extraction-verification reports, figures, and a referee-facing scope guide.

A Reproducibility, redshift-active verification, and data retention

A.1 Archive policy

The archive contains manuscript source, the redshift-active source implementation, canonical configurations, retained chain bundles, run metadata, post-processing scripts, figures, formal dependency papers, and machine-readable certificate outputs. Public Planck and Pantheon+/SH0ES likelihood payloads are not redistributed. They are identified by source instructions and, where used by the certificate, SHA-256 hashes. The small DESI DR1 reference vector and covariance are retained for deterministic audit reconstruction.

No custom observational data set is introduced by AASC. The additional numerical structure is the declared transport map, the executable penalty, and the certificate ledger.

A.2 Retained environment

The production environment is recorded as Python 3.13, Cobaya 3.6.2, CAMB 1.6.6, NumPy 2.3.5, SciPy 1.17.1, pandas 2.3.3, and matplotlib 3.10.9. The exact version ledger is retained in

```
computational_pipeline_v9_code_only/requirements_lock_retained.txt
```

A.3 Primary redshift-active source and configuration

The source implementation is retained in the ordinary pipeline tree, including:

```
computational_pipeline_v9_code_only/models/background.py
computational_pipeline_v9_code_only/inference/pantheon_cholesky.py
computational_pipeline_v9_code_only/official_likelihoods/desi_dr1_official.py
computational_pipeline_v9_code_only/external_likelihoods/aasc_official_joint.py
computational_pipeline_v9_code_only/scripts/validate_redshift_active_transport.py
```

The primary configuration is

```
configs/cobaya/aasc_planck2018_lite_native_deoverlap_redshift_active.yaml
```

with `redshift_active_transport: true`, `include_shoes: false`, $z_{\text{stop}} = 10$, $p = 4$, and $\lambda = 10^6$. Smoke-test and zero-correction validation configurations are retained alongside it.

The source validation is reproduced from `computational_pipeline_v9_code_only/` with

```
python scripts/validate_redshift_active_transport.py
```

and checks the zero-correction ΛCDM limit, independent redshift-shape effects, positivity, distance integration, and configuration flags.

A.4 Retained posterior ensembles

Run	Saved rows	Weighted samples	Maximum selected split-$\widehat{R}-1$
Primary redshift-active AASC	247,840	422,716	0.000984
Present-epoch de-overlapped AASC subcase	95,400	160,960	0.00247
Matched de-overlapped unrestricted ΛCDM	48,720	78,671	0.00350
Archived-stack unrestricted ΛCDM	52,920	85,175	0.00214
Archived-stack AASC	160,040	267,197	0.00312

Table 9: Retained multichain ensembles. Each row contains five independent production chain files.

The primary and de-overlapped comparison chain archives are stored at

```
results/redshift_active_aasc_multichain/chains_and_run_metadata.zip
results/deoverlap_aasc_multichain/chains_and_run_metadata.zip
results/deoverlap_lcdm_baseline_multichain/chains_and_run_metadata.zip
```

The primary redshift-active archive includes chain text files, input and resolved YAML, covariance matrices, checkpoints, progress files, and logs. The five production logs report sampling completion and convergence. PowerShell wrapper strings retained in some logs are launch-wrapper diagnostics; they are not failed-chain markers. The slim publication package retains the archived-stack posterior summaries, inventories, configurations, figures, and verification tables but omits their redundant historical full-chain ZIPs; those are preserved in the prior archival release.

A.5 Redshift-active ingest and posterior verification

The completed run was independently ingested without rerunning Cobaya. The retained assessment is under

```
results/redshift_active_assessment/
```

and includes chain inventory, posterior summaries, chain-level weighted means, split-$\widehat{R}$ diagnostics, redshift-dependent $\Delta H(z)$ summaries, best-sample rows, configuration flags, log scans, verification checks, and primary figures. The chain ZIP passed integrity testing and exactly five production chain files were identified.

A.6 Reproduce and validate the redshift-active certificate

After obtaining the public Pantheon+SH0ES and DESI inputs at the paths described in the code README, run from the project root:

```
python scripts/certificate/compute_redshift_active_aermax_jacobian.py \
  --project-root . --posterior-draws 24
python scripts/certificate/build_redshift_active_certificate_ledgers.py \
  --project-root .
python scripts/certificate/validate_redshift_active_certificate.py \
  --project-root .
```

The outputs are retained under

```
results/redshift_active_certificate/
```

and include the certificate scope, role-typing ledger, seven coordinate ledgers, evaluation points, singular values, kernel basis, standing projections, transport-subspace sensitivity, residual vector, certificate report, and a 45-check validation report.

The certificate scripts do not rerun the MCMC chains. They evaluate the declared source, observable, and standing maps at retained posterior points. The complete declared Jacobian is rank six/nullity zero at every retained audit point and certificate threshold; the transport subspace is rank three.

A.7 Nested present-epoch certificate and retained sensitivities

The earlier present-epoch certificate remains under

```
results/seven_coordinate_certificate/
```

as a nested subcase. The matched de-overlapped and archived-stack comparisons, archived-stack penalty diagnostics at $\lambda = 10^5$ and 10^7, and the historical importance-reweighting audit remain available for provenance and sensitivity analysis.

A.8 Fresh likelihood execution

After installing the public likelihood dependencies and fetching the raw public data, the primary run can be launched from `computational_pipeline_v9_code_only/` with independent output names for each chain, using

```
cd ../configs/cobaya
cobaya-run aasc_planck2018_lite_native_deoverlap_redshift_active.yaml
```

The matched de-overlapped baseline is executed with

```
cobaya-run ../configs/cobaya/lcdm_matched_native_deoverlap.yaml
```

A.9 Claimed and unclaimed extensions

The archive retains the source-supported redshift-active five-chain posterior, the redshift-active restricted seven-coordinate certificate, the nested present-epoch certificate, matched unrestricted baselines, archived-stack sensitivities, and the formal dependency bundle. It does not claim a nonparametric or arbitrary-function transport certificate, a full redshift-active multichain penalty profile, Bayesian evidence, a full Planck `clik/clipy` production run, or exhaustion of new-physics theorem objects.

B Matched-baseline and stack-comparison status

The matched unrestricted ΛCDM posterior uses the same de-overlapped observational ledger as the primary redshift-active AASC run, with the transport coefficients fixed to zero and $\lambda = 0$. It gives

$$H_0^{\Lambda\mathrm{CDM}} = 68.7185 \pm 0.3680\ \mathrm{km\,s^{-1}\,Mpc^{-1}},$$

while the primary redshift-active result is

$$H_{0,\mathrm{RA}}^{\mathrm{AASC}} = 68.7319 \pm 0.3675\ \mathrm{km\,s^{-1}\,Mpc^{-1}}.$$

The posterior-mean difference is $0.0134\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, negligible compared with either uncertainty. The comparison shows that the de-overlapped data ledger remains in the Planck/BAO-supported region and that the redshift-active transport sector is not activated.

The nested present-epoch AASC subcase gives $H_{0,\mathrm{PE}}^{\mathrm{AASC}} = 68.7382 \pm 0.3618$. Its agreement with the redshift-active posterior shows that making the full three-coefficient basis observable-facing does not induce migration toward SH0ES.

For the archived likelihood ledger,

$$H_{0,\mathrm{archived}}^{\mathrm{AASC}} = 69.2078 \pm 0.3519, \qquad H_{0,\mathrm{archived}}^{\Lambda\mathrm{CDM}} = 69.2156 \pm 0.3456.$$

The approximately $0.5\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ upward shift produced by the separate explicit SH0ES Gaussian factor occurs in both AASC and unrestricted ΛCDM. It is therefore a likelihood-ledger effect rather than activation of the transport correction.

The baseline role is posterior-locational and diagnostic. Supported conclusions are:

- the primary redshift-active AASC and matched ΛCDM posteriors occupy the same broad posterior region;
- all three redshift-active transport coefficients remain consistent with zero;
- the explicit SH0ES Gaussian factor produces the same modest upward displacement in the archived AASC and unrestricted posteriors.

No Bayesian evidence, Bayes factor, AIC/BIC analysis, or prior-volume-normalized model-selection preference is reported.

C Archived-stack penalty-weight sensitivity diagnostic

This appendix reports the retained penalty-weight sensitivity diagnostic for the archived likelihood stack, which includes the calibrated Pantheon+SH0ES covariance and the separate explicit SH0ES Gaussian summary factor. The archived five-chain ensemble uses

$$\lambda = 10^6,$$

while two additional single-chain diagnostic runs were retained at $\lambda = 10^5$ and 10^7. These runs predate the direct de-overlapped primary posterior and are retained as an archived-stack robustness check, not as a de-overlapped penalty profile.

The penalty is not a cosmological prior on H_0. It is the numerical mechanism enforcing the declared executable transport-closure representative. Smaller values of λ permit larger excursions in the regularized transport sector; larger values sharpen the hard-penalty limit of the executable representative. The implementation audit shows that, in the retained samples, $\mathcal{V}_{\mathrm{AASC}}^{\mathrm{exec}}$ is supported by the tensor/continuation regularization term because the other executable terms vanish.

Penalty weight	H_0^{AASC}	$\Delta_T(0)$	mean $\mathcal{V}_{\text{AASC}}^{\text{exec}}$	diagnostic convergence
10^5	69.249 ± 0.347	0.238 ± 0.171	4.41×10^{-5}	$R - 1 = 0.024$
10^6	69.208 ± 0.352	0.0267 ± 0.0588	3.18×10^{-6}	$\max(\widehat{R} - 1) = 0.00312$
10^7	69.216 ± 0.355	0.0020 ± 0.0185	2.91×10^{-7}	$R - 1 = 0.026$

Table 10: Archived-stack penalty-weight sensitivity. The 10^6 row is the archived five-chain AASC ensemble; the 10^5 and 10^7 rows are single-chain diagnostics.

The relaxed $\lambda = 10^5$ run permits a larger present-epoch correction, $\Delta_T(0) = 0.238 \pm 0.171$, and a larger mean executable violation. This remains far below the $\sim 3.8\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ shift required to reach the SH0ES central value from the archived posterior. The stricter $\lambda = 10^7$ run drives the correction closer to zero. Thus the archived posterior location is not produced by one isolated penalty weight.

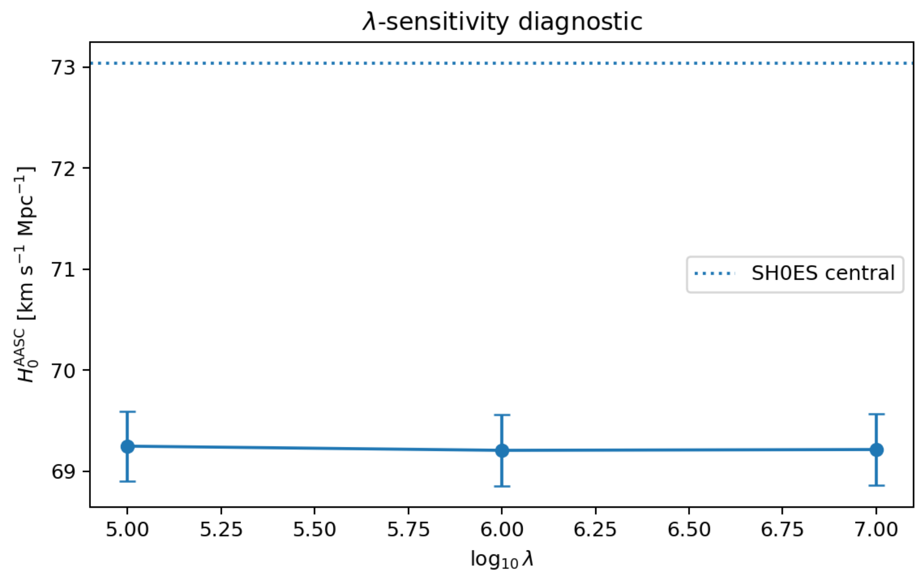


Figure 5: Archived-stack sensitivity of H_0^{AASC} to the penalty weight. The dotted horizontal line marks the SH0ES central value.

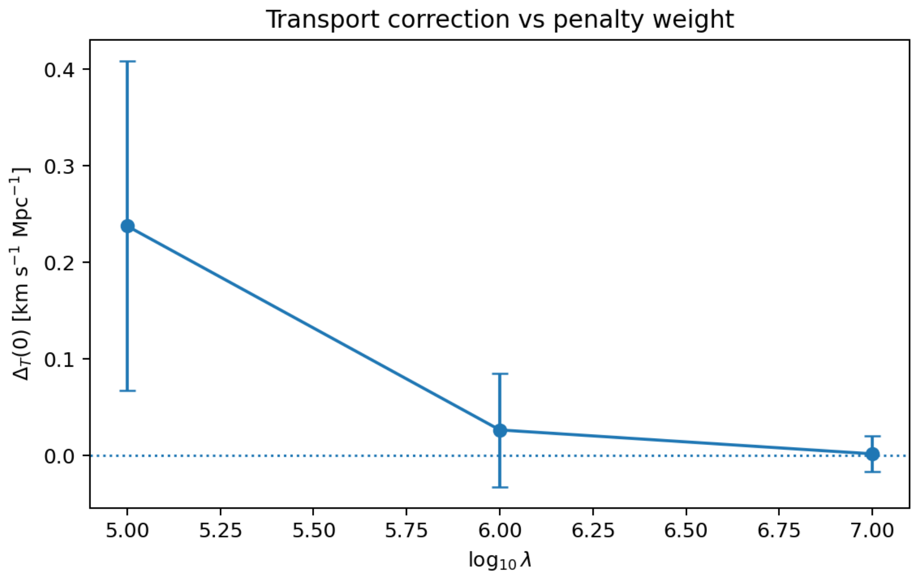


Figure 6: Archived-stack sensitivity of the present-epoch correction $\Delta_T(0)$ to the penalty weight. The hardening direction $10^6 \to 10^7$ drives the correction toward zero.

The primary redshift-active run is established independently in Section 18.4. A full redshift-active multichain λ-profile is not claimed. The retained archived scan nevertheless supplies a separate robustness check: relaxing the executable penalty over the retained range does not produce a SH0ES-scale transport correction.

D Restricted seven-coordinate certificate for the redshift-active theorem object

D.1 Certificate scope

This appendix certifies the finite redshift-active theorem object sampled by the primary five-chain run. Its scope is the native Planck validation anchor, DESI DR1 BAO, calibrated Pantheon+SH0ES covariance, no separate explicit SH0ES Gaussian summary factor, and the transport map

$$H_{\mathrm{AASC}}(z) = H_{\Lambda\mathrm{CDM}}(z) + \frac{dT_{a0} + dT_{a1}x + dT_{a2}x^2}{1 + (z/10)^4}, \qquad x = \frac{z}{1+z}.$$

The certificate is restricted to this declared data, model, transport, estimator, and tolerance ledger. It is not a certificate over arbitrary transport functions, unrestricted cosmological parameterizations, a full Planck `clik/clipy` production route, Bayesian evidence, or all new-physics extensions.

All three transport coefficients are typed as standing-bearing tensor witnesses *before* the Jacobian/SVD audit. This prospective role typing follows from the implemented redshift-active observable map: each coefficient changes the predicted SN and/or BAO observables while the anchor and boundary ledgers remain fixed.

D.2 Seven-coordinate ledger

Coordinate	Certificate content	Retained evidence	Residual status
Anchor A	Native Planck/CAMB early-output ledger, units, parameter envelope, and late-sector anchor suppression are fixed.	Primary YAML, source validation, hashes, and maximum prior-corner correction at recombination 3.89×10^{-7}, below the 10^{-6} tolerance.	$C_A = 0_{\text{tol}}$
Tensor W	$dT_{a0}, dT_{a1}, dT_{a2}$ are standing-bearing redshift-active tensor coordinates.	Role-typing ledger, source branches, nonzero observable derivatives, and rank-three transport subspace.	$C_T = 0_{\text{role/rank}}$
Quotient Q	The admitted observable class is the retained Planck anchor block plus covariance-whitened DESI BAO and calibrated Pantheon+SH0ES residuals.	Observable ledger, exact DESI/Pantheon covariance treatment, and full-rank declared Jacobian.	$C_Q = 0_{\text{map}}$
Cont. C	The early anchor is continued through the declared smooth redshift-active map with fixed $z_{\text{stop}} = 10$ and $p = 4$.	Source implementation, zero-correction limit, shape-effect tests, positivity tests, and distance-integral validation.	$C_C = 0_{\text{map}}$
Boundary B	Primary calibrated covariance and DESI records are fixed; the explicit SH0ES Gaussian factor is excluded.	Config flags, file hashes, data ledger, and archived-stack sensitivity kept separate.	$C_B = 0_{\text{ledger}}$
Estim. E	Cobaya/CAMB versions, priors, likelihood classes, convergence rules, and posterior summaries are declared.	Five chains, 247,840 rows, 422,716 weighted samples, and maximum selected split-$\widehat{R} - 1 = 0.000984$.	$C_E = 0_{\text{ledger}}$
Hidden remainder M	No observable-null direction changes standing-bearing content.	Rank-six/nullity-zero Jacobian at all retained audit points and declared thresholds; rank-three transport subspace.	$C_M = 0_{\text{num}}$

Table 11: Restricted seven-coordinate certificate for the declared redshift-active theorem object. The first six zeros are role, map, tolerance, or ledger zeros; C_M is the numerical Jacobian/SVD residual.

D.3 Coordinate charts and observable map

The local certificate chart is

$$\theta_{\text{cert}} = (H_0, \Omega_m, r_{\text{drag}}, dT_{a0}, dT_{a1}, dT_{a2}),$$

and the standing-content map is

$$S(\theta_{\text{cert}}) = (H_{0,\text{RA}}^{\text{AASC}}, dT_{a0}, dT_{a1}, dT_{a2}, \Omega_m, r_{\text{drag}}).$$

The admitted observable map Π_{RA} contains:

(i) the retained native-Planck/CAMB anchor-output block in $(H_0, \Omega_m, r_{\text{drag}})$;

(ii) exact covariance-whitened DESI DR1 BAO residuals evaluated with the redshift-active D_M, D_H, and D_V predictions;

(iii) exact covariance-whitened calibrated Pantheon+SH0ES residuals evaluated with the redshift-active luminosity distance and analytic absolute-magnitude marginalization.

The anchor block represents the retained early-output ledger. It is not claimed to be a raw CMB-likelihood response Jacobian.

D.4 Jacobian/SVD hidden-remainder audit

The linear hidden-remainder residual is

$$C_M = \|J_S P_{\ker J_\Pi}\|^2_{\mathrm{HS}} .$$

The Jacobians are evaluated by recorded central finite differences at the posterior mean, the minimum-χ^2 retained point, and 24 deterministic weighted posterior draws. Singular-value thresholds 10^{-6} and 10^{-5}, relative to the largest singular value, define the certificate kernel. A stricter 10^{-4} threshold is retained only as a conditioning stress diagnostic.

At every retained point and both certificate thresholds, the full declared map satisfies

$$\operatorname{rank} J_\Pi = 6, \qquad \dim\ker J_\Pi = 0.$$

The three-column transport sub-Jacobian also satisfies

$$\operatorname{rank} J_{\Pi,(dT_{a0},dT_{a1},dT_{a2})} = 3.$$

Therefore

$$P_{\ker J_\Pi} = 0, \qquad J_S P_{\ker J_\Pi} = 0, \qquad C_M = 0.$$

The smallest relative singular value across the retained points is 3.365×10^{-3} for the full declared map and 8.843×10^{-5} for the late-sector map. The late-sector map is full rank at the declared thresholds, although its larger condition number and failure under the deliberately stricter 10^{-4} stress threshold are retained as conditioning diagnostics. They do not alter the certificate defined at 10^{-6} and 10^{-5}.

D.5 Residual vector and validation status

The certified residual vector is

$$(C_A, C_T, C_Q, C_C, C_B, C_E, C_M) = (0, 0, 0, 0, 0, 0, 0) \tag{12}$$

relative to the declared modes and tolerances. The machine-readable validator passes 45 of 45 checks, including source presence, zero-correction and shape-effect tests, configuration flags, chain integrity, convergence, prospective role typing, full Jacobian rank, rank-three transport sensitivity, anchor suppression, and all seven residual entries.

The certificate files are retained under

`results/redshift_active_certificate/`

and include the scope declaration, role-typing ledger, seven coordinate ledgers, Jacobian evaluation points, singular values, kernel basis, projection norms, transport-subspace sensitivity, residual vector, certificate report, and validation report.

D.6 Relationship to the present-epoch certificate

The earlier present-epoch certificate remains valid as a nested restricted object. In that object, only dT_{a0} entered the late observables, while dT_{a1} and dT_{a2} served as continuation-enforcement coordinates. In the redshift-active object, the theorem object changes: all three coefficients enter the physical observable map and are therefore promoted prospectively into the tensor-witness ledger. The role change is a theorem-object change, not a post-hoc response to the Jacobian result.

D.7 Exact claim boundary

The certificate upgrades the primary result to a certificate-bearing redshift-active finite theorem object. Within that object, the absence of a SH0ES-scale correction is not attributable to anchor drift, tensor omission, quotient mismatch, continuation repair, boundary-record movement, estimator selection, or an observable-invisible standing-bearing direction.

The certificate does not establish that every conceivable redshift function, cosmological extension, data ledger, or physical mechanism is inadmissible. Such alternatives define new theorem objects and require their own observable maps, role ledgers, posterior sampling, and certificates.

E Referee guide: scope, status, and common misreadings

This guide records the exact status of the primary redshift-active result.

"The executable penalty is the full seven-residual theorem."
No. $\mathcal{V}^{\mathrm{form}}_{\mathrm{AASC}}$ and $\mathcal{V}^{\mathrm{exec}}_{\mathrm{AASC}}$ are distinct. The chain samples the executable representative; the separate role, ledger, observable-map, and Jacobian audits supply the restricted seven-coordinate certificate.

"A small $V^{\mathrm{exec}}_{\mathrm{AASC}}$ alone proves closure."
No. The scalar penalty is part of the sampled objective. Formal closure of the finite theorem object is established by the machine-readable seven-coordinate certificate, not inferred from the scalar penalty alone.

"Only dT_{a0} is observable."
That statement applies to the nested present-epoch subcase, not the primary redshift-active object. In the primary object, $dT_{a0}, dT_{a1}, dT_{a2}$ all enter the SN and DESI BAO predictions. The redshift-active transport sub-Jacobian has rank three at every retained audit point.

"The transport coefficients were called standing-bearing only after the Jacobian was examined."
No. Their role typing is fixed prospectively by the redshift-active source map and certificate scope. The Jacobian audit is then performed with all three coefficients in the standing-content chart.

"The AER-Max audit fails because there may be a hidden shape direction."
No hidden standing-bearing direction is detected at the declared thresholds. The full certificate Jacobian is rank six with nullity zero, and the three-dimensional transport subspace is rank three at all retained audit points. Hence $C_M = 0$.

"The late sector alone is necessarily singular."
Not in the redshift-active object at the declared certificate thresholds. The late-sector Jacobian is full rank, although less well conditioned than the full early/late map. A stricter 10^{-4} stress threshold is reported as a conditioning diagnostic, not as the certificate definition.

“The certificate is universal.”

No. It is restricted to the native Planck validation, DESI DR1, calibrated Pantheon+SH0ES covariance, de-overlapped three-coefficient transport object. Arbitrary transport functions, different data ledgers, additional state variables, or new dynamics require new certificates.

“The redshift-active result is only a smoke test.”

No. The source validation and smoke tests precede a completed five-chain production posterior with 247,840 saved rows, 422,716 weighted samples, and maximum selected split-$\widehat{R} - 1 = 0.000984$.

“AASC beats unrestricted ΛCDM.”

Not claimed. The matched de-overlapped ΛCDM posterior occupies essentially the same H_0 region. No Bayesian evidence, Bayes factor, AIC/BIC, or prior-volume-normalized preference is reported.

“The paper merely failed to find an arbitrary correction.”

No. The declared transport channel is explicitly constructed, prospectively role-typed, coupled to the retained observables, and closed by the seven-coordinate certificate. The stronger result is that the posterior does not occupy this admissible channel at the SH0ES scale.

“The theorem says that nature contains two unrelated expansion phenomena.”

No. The theorem is conditional and relation-specific. It states that the early anchor and the SH0ES-scale late determination do not satisfy the certified same-parameter transport relation within the declared redshift-active theorem object.

“The paper rejects SH0ES.”

No. The SH0ES distance is a reference diagnostic. The calibrated Pantheon+SH0ES covariance remains in the primary likelihood, while the separate explicit SH0ES Gaussian factor is excluded.

“The native Planck route is a full production `clik/clipy` result.”

No. It is the retained native validation route. Full production-likelihood replication is a distinct robustness target.

The supported endpoint is therefore precise: the source-supported redshift-active three-coefficient theorem object is sampled by a converged five-chain posterior and clears its restricted seven-coordinate certificate. Conditional on the imported AASC results, the early anchor and the SH0ES-scale late determination do not stand in the certified same-parameter transport relation within that object. Broader all-basis, evidence-based, SH0ES-rejection, or production-Planck claims are not made.

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